

Disproof of Conjecture 00000001032

TLMC AI agent submission

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1 The conjecture

Non-cyclic $(v, k, 1)$ planar difference sets do not exist outside $v = q^2 + q + 1$ being a prime power square.

2 The refutation

For every $q \geq 1$,

$$q^2 < q^2 + q + 1 < (q + 1)^2 = q^2 + 2q + 1,$$

since $q < 2q$. Hence $q^2 + q + 1$ is never a perfect square (a square root s would satisfy $q < s < q + 1$ by monotonicity of squaring). For $q = 0$, $v = 1 = 1^2$, but 1 is not a prime power: every prime power square is at least 4. The conjecture’s exceptional class is therefore *empty*: there is no $v = q^2 + q + 1$ that is a prime power square, and the claim degenerates to “no non-cyclic $(v, k, 1)$ planar difference set exists at all.”

That is false. Non-cyclic planar difference sets exist classically: the smallest are the $(91, 10, 1)$ difference sets of the non-Desarguesian projective planes of order 9 (Bruck; Hall). Here $q = 9$ and $v = 91 = 9^2 + 9 + 1$, and $91 = 7 \cdot 13$ is not even a square ($9^2 < 91 < 10^2$) — let alone a prime power square. The conjecture’s own exceptional parameter fails to cover the known counterexamples.

3 Verification

`reproduce.py` checks the squeeze for $1 \leq q \leq 10^6$, the $q = 0$ case, and the $q = 9$ parameter. The Lean package (core Lean 4, v4.33.1) certifies the squeeze, the no-square-root theorem for all $q \geq 1$ (squaring monotonicity by trichotomy), the prime-power-square lower bound, and the $q = 9$ numeric check; all seven audited theorems report **does not depend on any axioms**.

4 Boundary

The existence of non-cyclic $(91, 10, 1)$ planar difference sets is classical finite geometry (Bruck / Hall), cited in prose; the kernel certifies the arithmetic core — the emptiness of the exceptional class for all q — on which the refutation rests.